

A PRELIMINARY MINI PROJECT REPORT ON
“BookScape: An Online Bookstore Web Application using MERN Stack”

**SUBMITTED TOWARDS THE PARTIAL FULFILLMENT OF
THE REQUIREMENTS OF**

BACHELOR OF TECHNOLOGY (Third Year B. Tech.)

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PIMPRI CHINCHWAD COLLEGE OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING,

CERTIFICATE

This is to certify that, the mini project entitled
“BookScape: An Online Bookstore Web Application using MERN Stack”
is successfully carried out as a mini project for Full Stack Development
Laboratory and successfully submitted by following students of “PCET's
Pimpri Chinchwad College of Engineering, Nigdi, Pune-44”.

Under the guidance of Prof. Manisha Pawar

In the partial fulfillment of the requirements for the Third Year B. Tech. (Computer
Engineering)

Prof. Manisha Pawar
Project Guide

Abstract

The **BookScape** project is a full-stack web application developed using the MERN stack, which includes MongoDB, Express.js, React, and Node.js. The primary objective of this system is to provide a user-friendly and efficient platform for buying and browsing books online.

In today's digital era, users prefer online platforms for convenience, accessibility, and variety. BookScape addresses these needs by allowing users to explore a wide range of books, search for specific titles, and manage their cart with ease. The system also includes an administrative interface that enables efficient management of book data such as adding, updating, and deleting records.

The project demonstrates the integration of frontend and backend technologies, efficient database handling, and RESTful API communication. Overall, BookScape provides a scalable solution for modern e-commerce systems and serves as a practical implementation of full-stack development concepts.

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Chapter 1: Introduction

1.1 Problem Statement

In traditional bookstore systems, users are required to physically visit stores to search for and purchase books. This process is time-consuming and limits access to a wide variety of books. Additionally, users often face difficulty in finding specific titles, comparing prices, and accessing updated information about book availability.

Existing online platforms, although functional, are often overloaded with features and complex user interfaces, which may confuse users and reduce usability. There is a need for a simple, efficient, and user-friendly platform that allows users to browse, search, and manage book purchases seamlessly.

The problem addressed in this project is to design and develop a streamlined online bookstore system that enhances user experience while maintaining efficiency and scalability.

1.2 Project Idea

The idea behind **BookScape** is to develop a modern web-based bookstore application that provides a smooth and intuitive user experience. The system is designed to allow users to browse books, search for specific titles or categories, and view detailed information about each book, including price, author, and description.

The application also includes a cart feature where users can add or remove books before making a purchase. In addition, an admin module is incorporated to manage the inventory efficiently by allowing the addition, modification, and deletion of books.

By using the MERN stack, the project ensures seamless communication between the frontend and backend, making the application dynamic, responsive, and scalable.

1.3 Motivation

The motivation behind developing this project comes from the increasing demand for online shopping platforms and the growing importance of digital accessibility. With the advancement of web technologies, users expect fast, reliable, and visually appealing applications.

Another key motivation is to gain practical experience in full-stack development. This project provides hands-on exposure to real-world technologies such as React for frontend development, Node.js and Express.js for backend services, and MongoDB for database management.

Additionally, the project helps in understanding how different components of a web application interact with each other, which is essential for building scalable and efficient systems in the software industry.

1.4 Scope

The scope of the BookNest project includes the development of a complete web-based bookstore system with essential functionalities such as user authentication, book browsing, search and filtering, and cart management. The system is designed to provide a responsive and user-friendly interface that works across different devices.

The current implementation focuses on core features required for an online bookstore. However, the system can be further enhanced in the future by integrating advanced features such as online payment gateways, recommendation systems, user reviews, and order tracking.

Thus, the project provides a strong foundation that can be extended into a fully functional commercial application.

1.5 Literature Survey / Requirement Analysis

A study of existing online bookstore platforms such as Amazon and Flipkart reveals that while these systems provide a wide range of features, they often have complex interfaces and may overwhelm users. These platforms are designed for large-scale operations and may not always prioritize simplicity and ease of use.

Based on this analysis, the requirements for the proposed system were identified. The system should provide a clean and intuitive interface, fast performance, and efficient data handling. It should allow users to easily search for books, view details, and manage their cart without unnecessary complexity.

Functional requirements include user authentication, book management, and cart functionality, while non-functional requirements include performance, scalability, and usability. The proposed system aims to address these requirements effectively.

Chapter 2: Project Design

2.1 Hardware Requirements

The development and execution of the BookNest application require basic hardware resources. A system with an Intel i3 processor or higher is sufficient to handle the development environment and application runtime. A minimum of 4GB RAM is required to ensure smooth performance while running development tools and servers.

Additionally, sufficient storage space is necessary to install required software, dependencies, and project files. These hardware requirements ensure that the application can be developed and tested efficiently without performance issues.

2.2 Software Requirements

The BookNest application is built using modern web development technologies. The frontend is developed using React, which provides a dynamic and responsive user interface. The backend is implemented using Node.js and Express.js, which handle server-side logic and API requests.

MongoDB is used as the database to store book and user data in a flexible and scalable manner. Development tools such as Visual Studio Code are used for coding, while Postman is used for testing APIs.

These technologies together form the MERN stack, which is widely used for building full-stack web applications.

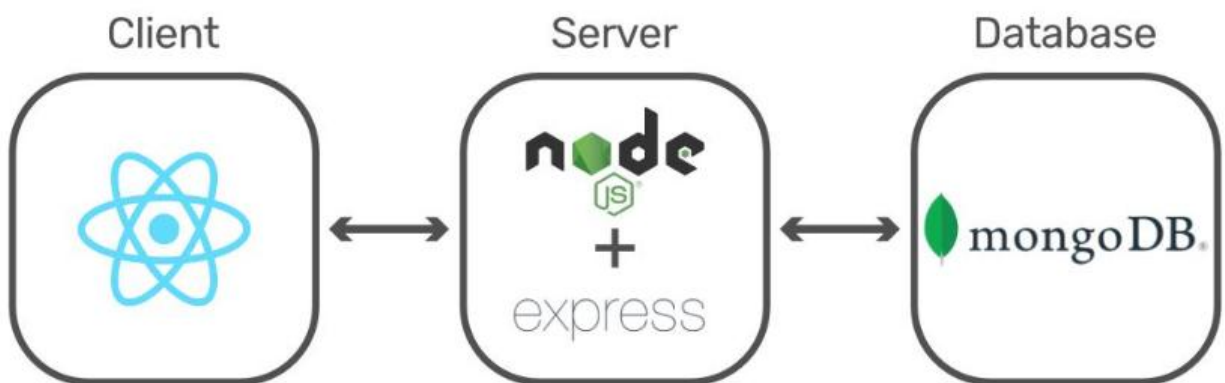


fig. 2.2.(a) - MERN Stack

2.3 Dataset Design

The dataset for the BookNest application consists of book-related information stored in a MongoDB database. Each book record includes attributes such as book ID, title, author, price, category, description, and image URL.

The database is designed in a structured manner to ensure efficient storage and retrieval of data. Proper indexing and schema design are used to optimize performance and ensure scalability as the number of records increases.

```
_id: ObjectId('69e65c6971e81fff667427e0')  
title : "The Psychology of Money"  
author : "Morgan Housel"  
price : 399  
image : "https://m.media-amazon.com/images/I/71g2ednj0JL.jpg"  
rating : "4.5"
```

fig.2.3(a)-Database Design

2.4 Hours Estimation

The development of the BookNest project was planned and executed in phases. Initial hours were allocated for requirement analysis and system design to understand the problem and define the solution. This was followed by frontend and backend development, where the core functionalities of the system were implemented.

Additional time was spent on integrating different components, testing the application, and fixing bugs. Proper time management ensured that the project was completed within the given timeframe while maintaining quality.

Chapter 3: Module Description

3.1 Module Description

The BookNest system is divided into multiple modules to ensure modularity and maintainability. The user module handles user authentication and allows users to browse and interact with the system. The admin module provides functionalities for managing books, including adding, updating, and deleting records.

The book module is responsible for displaying book information and handling search and filtering operations. The cart module allows users to add books to their cart, remove items, and manage their selections before purchase. Each module is designed to perform specific tasks, and together they form a complete and efficient system.

3.2 Block Diagram Explanation

The system architecture follows a client-server model. The user interacts with the frontend, which is built using React. The frontend communicates with the backend server developed using Node.js and Express.js through API calls.

The backend processes these requests and interacts with the MongoDB database to store or retrieve data. The response is then sent back to the frontend and displayed to the user. This architecture ensures a clear separation of concerns, making the system scalable and easy to maintain.

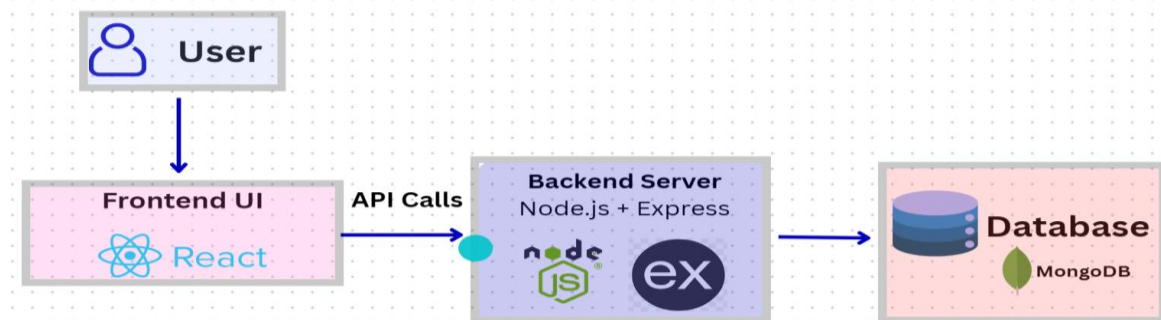


fig.3.2(a) - Block Architecture Diagram of BookNest

3.3 Comparative Study

A comparison between existing systems and the proposed BookNest system shows that while existing platforms (BookMandee, ClantKart) offer a wide range of features, they often lack simplicity. BookNest focuses on providing a clean and intuitive interface, making it easier for users to navigate and perform tasks.

The system is optimized for performance and provides faster response times due to its lightweight design. Additionally, it allows greater customization and flexibility compared to traditional systems.

Chapter 4: Results & Discussion

4.1 Source Code

The source code of the BookScape application is organized into frontend and backend directories. The frontend contains React components, while the backend includes server logic and API routes.

The code is modular and well-structured, making it easy to understand and maintain. Proper coding practices and documentation have been followed to ensure readability and scalability.

Following is the Repository of Source Code:

<https://github.com/Pradnya1227/BookStoreApp.git>

4.2 Screenshots Including GUI

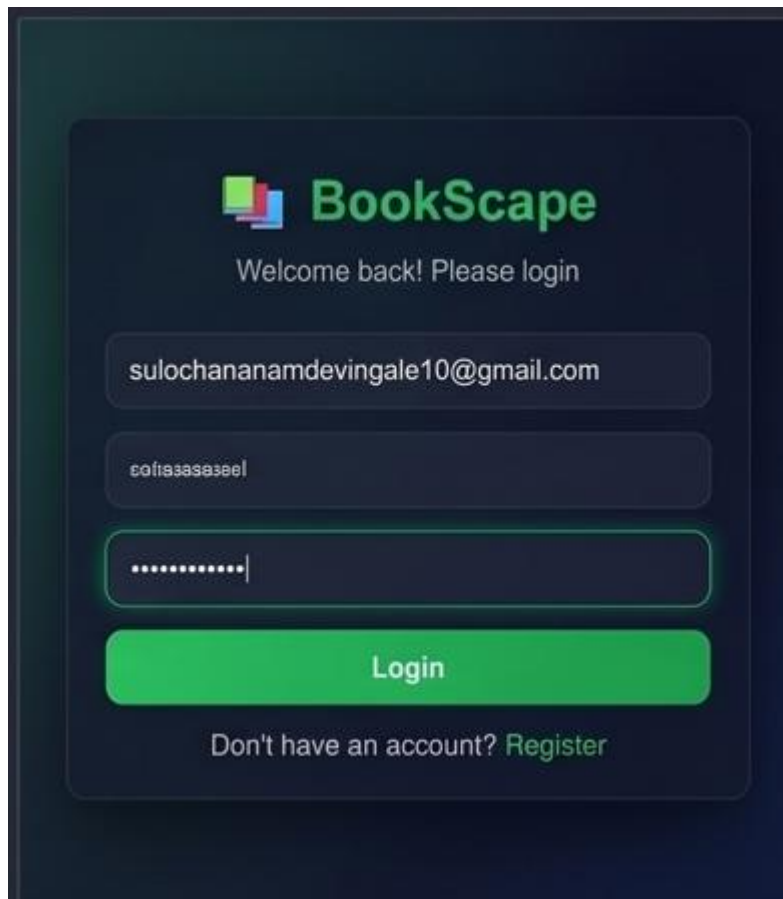
The graphical user interface of the application includes multiple screens such as the home page, login page, book listing page, Wishlist page and cart page. Each screen is designed with a focus on simplicity and usability.

The interface is responsive and adapts to different screen sizes, ensuring a consistent user experience across devices. Screenshots of these interfaces demonstrate the functionality and design of the system.

4.2.1 Login & Registration

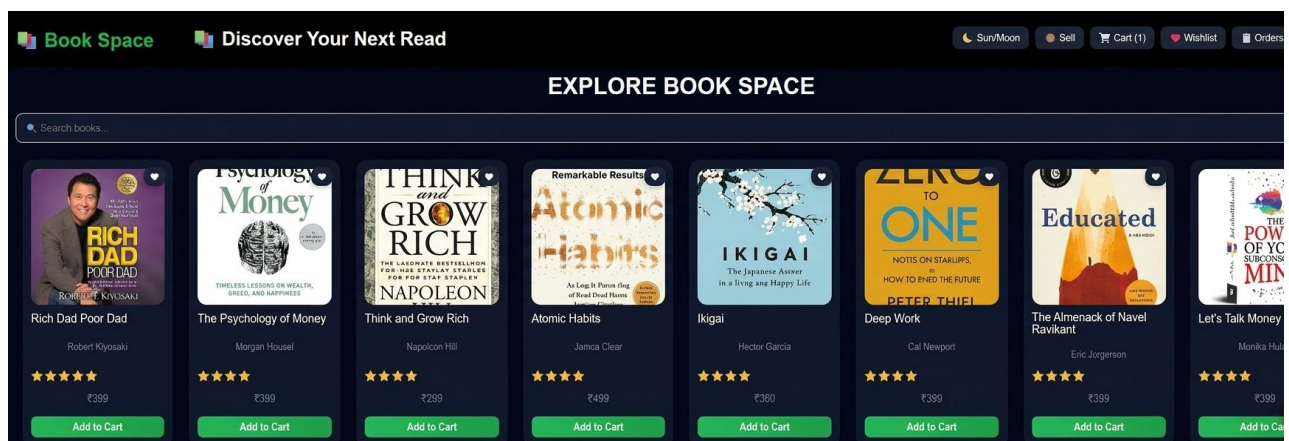
Provides secure user authentication with clean input fields and a centered card layout. Allows smooth switching between login and register with a modern UI design.





4.2.2 Home Page (Book Listing)

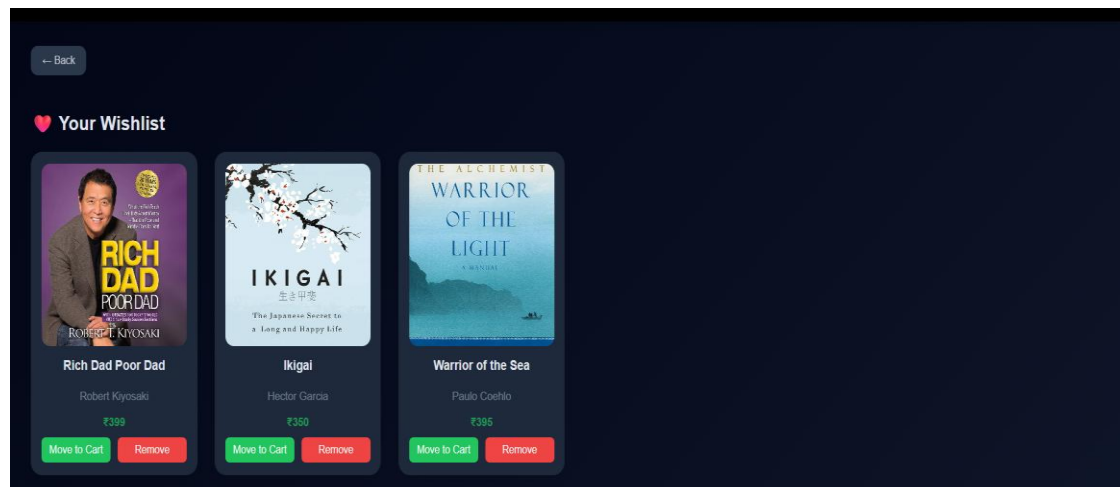
Displays books in a responsive grid with image, title, rating, and price. Includes search and price filter features for easy navigation.





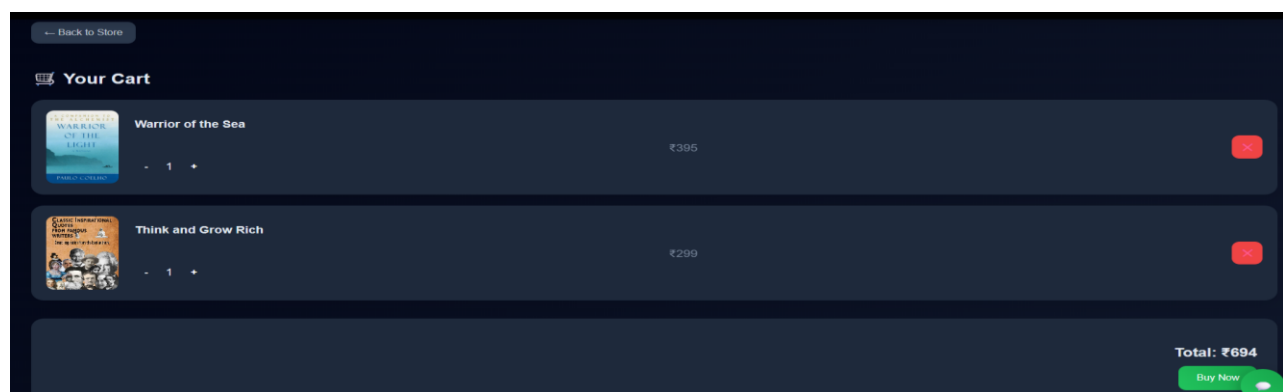
4.2.3 Wishlist

Stores user-selected books for future reference in a grid layout. Provides options to remove items or move them to the cart.



4.2.4 Cart Page

Displays selected items with quantity controls and total price calculation. Allows users to remove items and proceed to checkout.



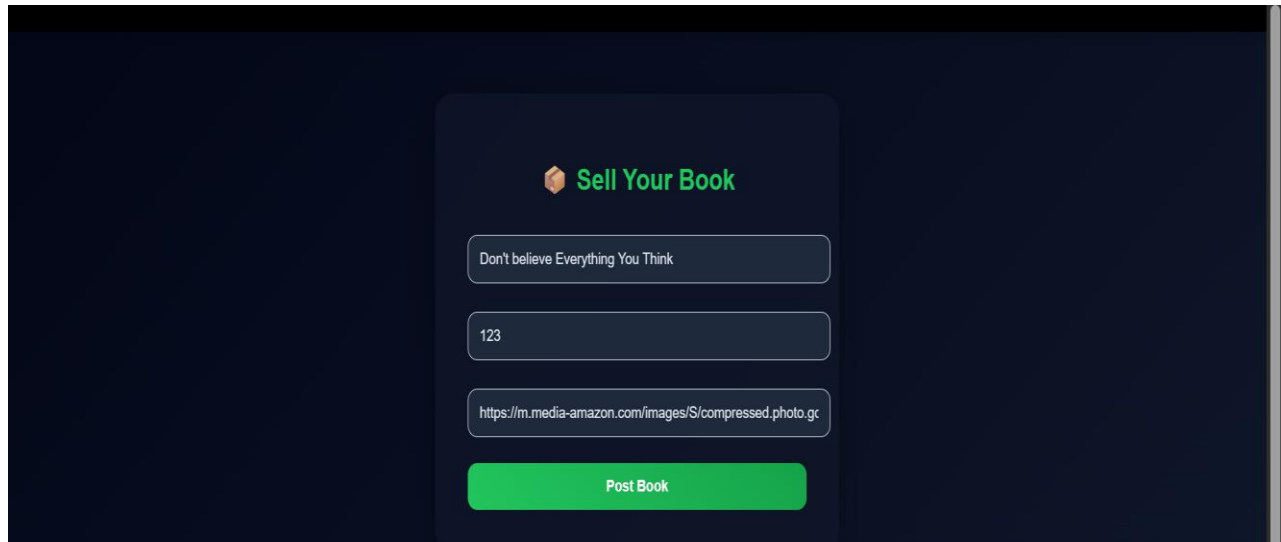
4.2.5 Order History

Shows all past orders with order ID, date, and total amount. Helps users track their purchases easily.



4.2.6 Sell Book

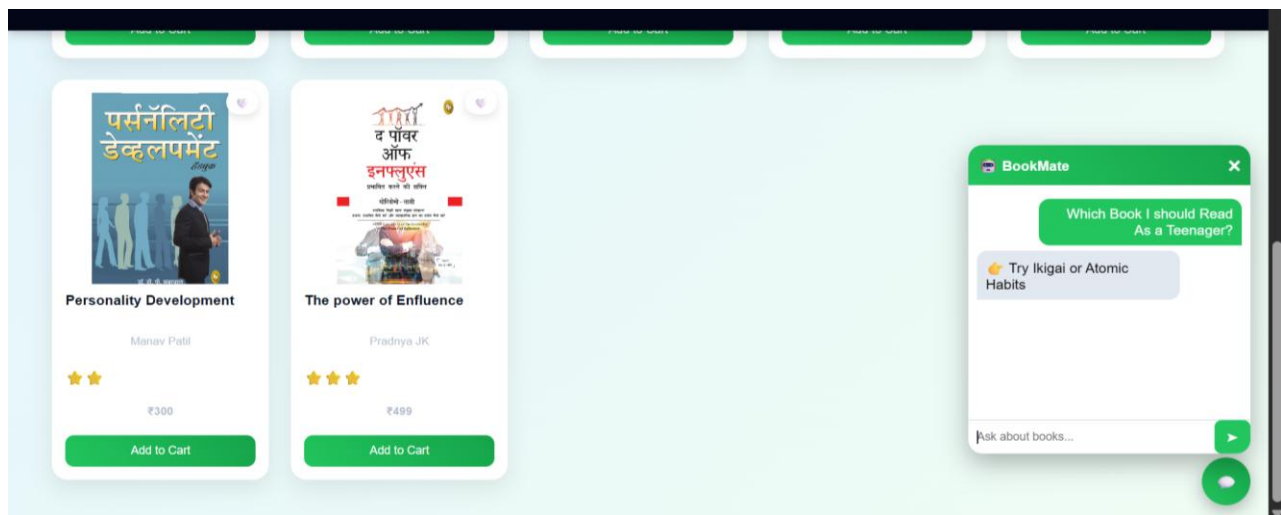
Allows users to add and list their own books for selling. Includes simple input fields for title, price, and image.



The screenshot shows a dark-themed interface for selling a book. At the top, there is a green icon of a book and the text "Sell Your Book". Below this, there are three input fields: the first contains the text "Don't believe Everything You Think", the second contains the number "123", and the third contains a URL "https://m.media-amazon.com/images/S/compressed.photo.gc". At the bottom of the form is a green button labeled "Post Book".

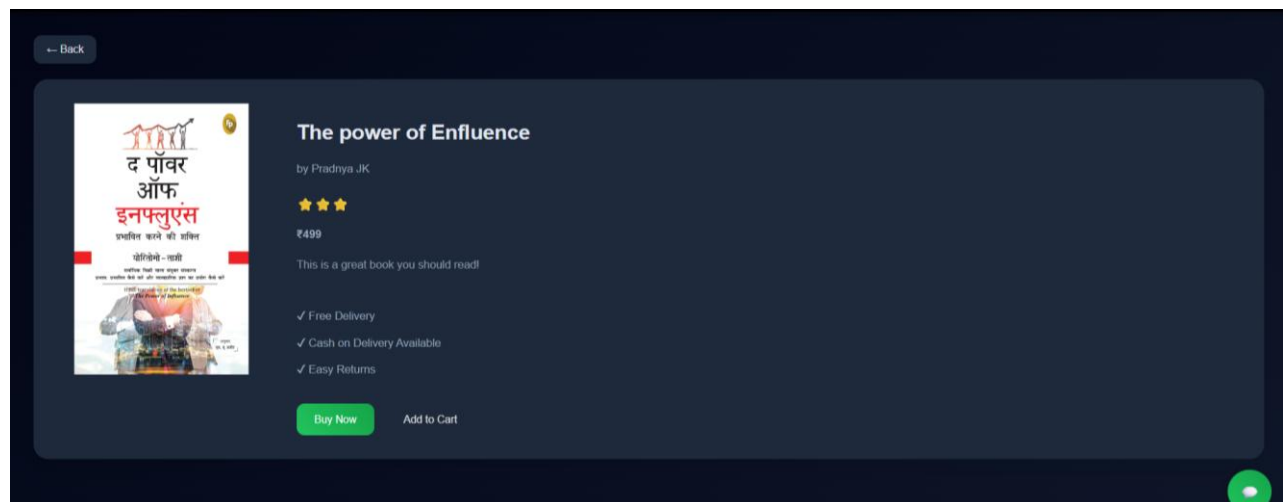
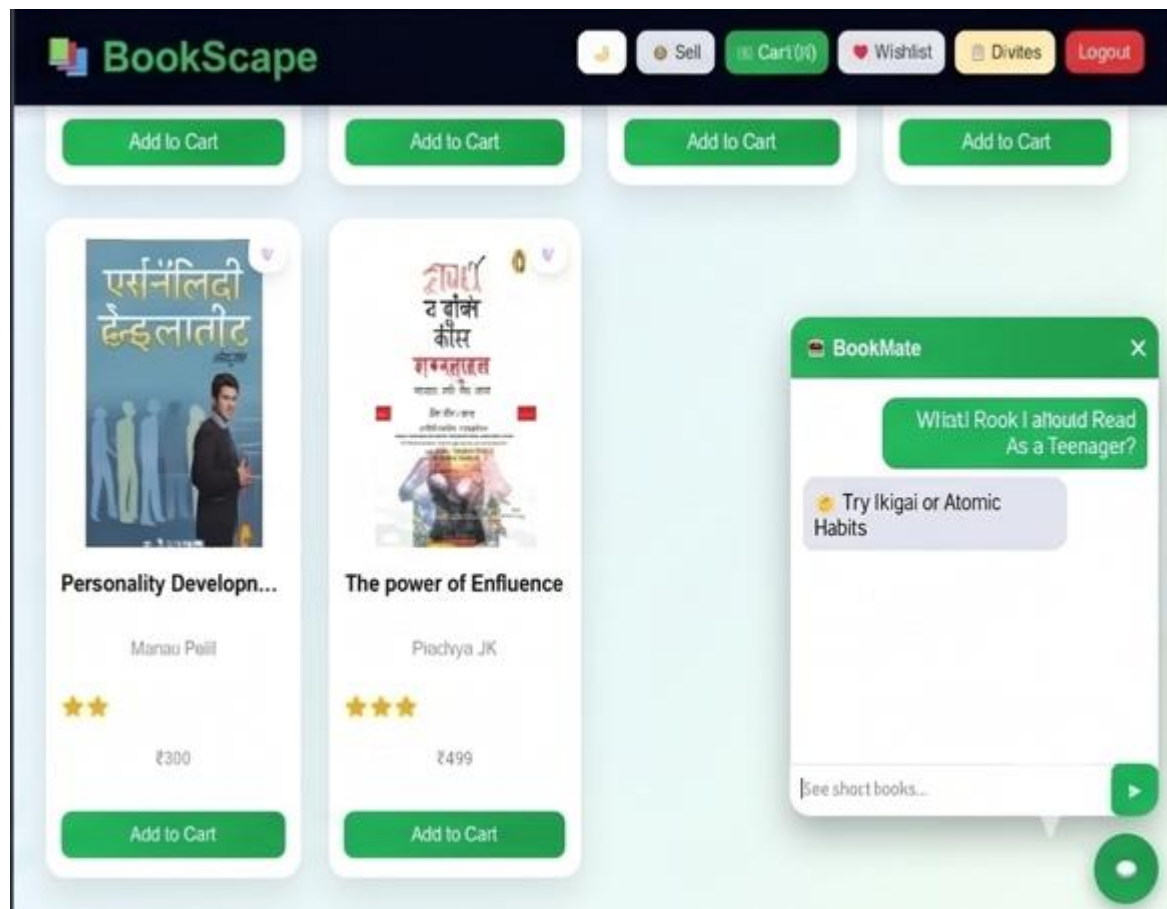
4.2.7 Chatbot

Provides basic book suggestions based on user queries. Enhances user interaction and guidance within the platform.



4.2.8 Product Details

Shows detailed information about the selected book with image, price, and description. Provides actions like **Buy Now** and **Add to Cart**.



4.2.9 Data Collection

localhost:27017 > bookstore > books

Documents 11 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) ✚

[+ ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT DATA](#) [EXPORT CODE](#)

```
_id: ObjectId('69e65bfb71e81fff667427de')
title: "Rich Dad Poor Dad"
author: "Robert Kiyosaki"
price: 399
image: "https://m.media-amazon.com/images/I/814XbqXAz-L._UF894,1000_QL80_.jpg"
rating: "5"
```

```
_id: ObjectId('69e65c6971e81fff667427e0')
title: "The Psychology of Money"
author: "Morgan Housel"
price: 399
image: "https://m.media-amazon.com/images/I/71g2ednj0JL.jpg"
rating: "4.5"
```

```
_id: ObjectId('69e65c6971e81fff667427e1')
title: "Think and Grow Rich"
author: "Napoleon Hill"
price: 299
image: "https://m.media-amazon.com/images/I/71UypkUjStL.jpg"
rating: "4.5"
```

localhost:27017 > bookstore > users

Documents 2 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) ✚

[+ ADD DATA](#) [UPDATE](#) [DELETE](#) [EXPORT DATA](#) [EXPORT CODE](#)

```
_id: ObjectId('69e65c1af0203f62dc4a4fa5')
name: "Shreeya Patil"
email: "shreeya@45"
password: "$2b$10$yB9gm4HvQTwAgRl/VjRUfeYFwIPur2jCglNW90vgbndkCdNErYBK2"
__v: 0
```

```
_id: ObjectId('69e67b8cf0203f62dc4a4fa6')
name: "Anuradha Kadam"
email: "Anuradha@21"
password: "$2b$10$cN3kIr5KYqIB1u17E4B5wOSwXlL5lVDUZSpxirvzHAI0qV4jmsoT6"
__v: 0
```


4.3 Test Cases

Various test cases were executed to ensure the correctness and reliability of the system. These include testing user authentication, book management, search functionality, and cart operations.

The results show that the system performs as expected under different scenarios. All major functionalities were tested successfully, indicating that the application is stable and ready for deployment.

Test Case ID	Description	Input / Action	Expected Output	Result
TC01	User Login	Enter valid email & password and click Login	User should be successfully logged in and redirected to home page	Pass
TC02	Add Book to Cart	Click "Add to Cart" on any book	Book should be added to cart and cart count should increase	Pass
TC03	Place Order (COD)	Click "Buy Now" → Select Cash on Delivery	Order should be placed successfully and shown in Order History	Pass

Chapter 5: Conclusion

The BookNest project successfully illustrates the development of a full-stack web application using the MERN stack. The system provides an efficient and user-friendly platform for browsing and managing books online.

From this project, key concepts of frontend and backend development, database management, and API integration were implemented and understood in a clear way. The application meets its defined objectives and provides a strong foundation for future enhancements.

With additional features such as payment integration and recommendation systems, the project can be further developed into a fully functional commercial product.

References

1. *React Documentation*, Available on: <https://react.dev/>
2. *Node.js Documentation*, Available on: <https://nodejs.org/>
3. *Express.js Documentation*, Available on: <https://expressjs.com/>
4. *MongoDB Documentation*, Available on: <https://www.mongodb.com/docs/>
5. *Web Development with Node.js and Express*, O'Reilly Media, 2019
6. *Learning React*, O'Reilly Media, 2020.

Annexure

A. Mapping With SDG Goal

SDG: Quality Education (SDG 4)

The BookSpace platform aligns primarily with SDG 4 (Quality Education) by promoting access to books and knowledge. Additionally, it supports SDG 8 (Decent Work and Economic Growth) through its book selling feature and SDG 9 (Industry, Innovation and Infrastructure) by leveraging modern web technologies.